Regression
Linear Regression and
Multiple linear regression

Lecture Outline

- **Regression Techniques**
  - **Linear regression**
  - **Non-linear Regression**
- **Typical applications of regression**
- **Linear regression**
  - **Key Concepts of Linear Regression**
  - **Typical simple regression model of Linear Regression**
  - **Evaluating Linear Regression Model**
- **Non-linear Regression**
- **Linear Regression in Python**

Regression Techniques

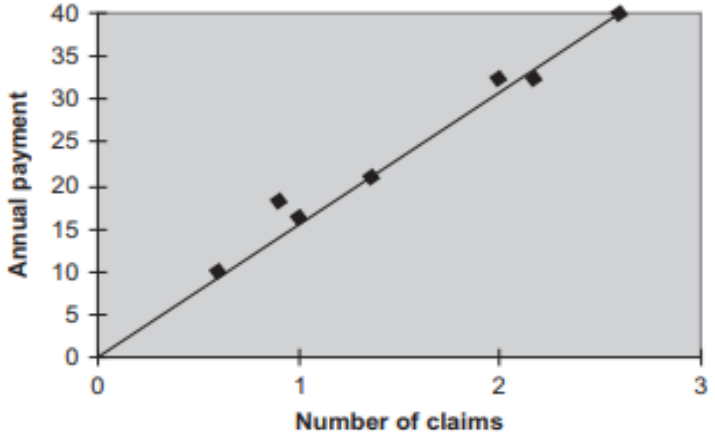
- Regression is the technique most often used to create **data models** in traditional statistics.
- It is a statistical modeling technique that examines the relation between a **dependent** variable and one or more **independent** variables.
- There are three main types:
  - linear regression, for data whose distribution tends to adjust to a straight line;
  - non-linear regression, for data whose tendency adjusts to a curve; and
  - logistic regression, for data models whose output is a binary type.

Linear regression

- Linear regression is one of the oldest predictive methodologies, it the most easily method for demonstrating function fitting.
- The basic idea is to come up with a function that explains and **predicts** the value of the **target variable** when given the values of the **predictor variables.**
- In **linear regression**, the primary objective is to predict numerical features such as real estate or stock prices, temperature, exam marks, and sales revenue. This method is particularly effective when dealing with continuous predictor and target variables.

Linear Regression .....

- Both the dependent and the independent variables have to be numerical. Categorical variables such as gender or district of residence must be recoded as binary variables (two categories) or as numerical.
- Linear regression analyzes the relation between two variables, X and Y, and tries to find the best straight line that passes through the data, as seen in below Figure



Key Concepts of Linear Regression :

Continuous Variables:

1. **Predictor Variable:** The input variable used for prediction.
2. **Target Variable:** The variable we aim to predict.

Straight Line Relationship:

1. In linear regression, a straight line is 'fitted' to represent the relationship between predictor and target variables.

Statistical Technique: Least Squares Method:

1. The fitting process employs the statistical concept of the least squares method.
2. **Objective:** Minimize the sum of squared differences between observed and predicted values.

Examples of Linear Regression

- **Predicting Stock Prices : As an example, consider predicting stock prices using linear regression:**
  - ❑ **Predictor Variables: Historical stock data.**
  - ❑ **Target Variable: Future stock prices.**
  - ❑ **Visual representation of the fitted line illustrates the predictive relationship.**
- **Predict the total volume of purchases : Let's say you work for an e-commerce company, and your task is to predict how much a client might spend in a quarter. Here, we have two types of variables:**
  - ❑ **Predictor Variable (Independent Variables): These are the factors we believe influence the total volume of purchases. In this case:**
    - Age of the client
    - Income of the client
    - Socioeconomic indicator, like the district of residence
  - ❑ **Target Variable (Dependent Variable): This is what we want to predict – the total volume of purchases made by a client in a given quarter.**

Simple Regression model of Linear Regression

- **Linear regression** is the fact that there is a significant correlation between two variables to make **predictions** about one variable based on the knowledge of the other.
- The straight line of linear regression model is expressed in the form:

$$Y = (a + bX) + \varepsilon$$

- **Residual ε** is the distance between the predicted point (on the regression line) and the actual point
- The objective of linear regression is to find the **line** that best predicts Y given X.

Computing a and b for a set of X and Y values.

Steps to Calculate 'a' and 'b':

1. Determine the Mean of X and Y:

$$\overline{X} = \frac{\sum X}{n} \qquad \overline{Y} = \frac{\sum Y}{n}$$

2. Calculate 'b' (Slope):

$$\boldsymbol{b} = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_i (X_i - \overline{X})^2}$$

3. Calculate 'a' (Intercept): $a = \left(\overline{Y} - b\overline{X}\right)$

Evaluating Linear Regression Model

Once we've built a Linear Regression (LR) model, it's crucial to evaluate its performance. Let's explore key evaluation methods to ensure our model is effective.

1. **Sum of Squared Errors (SSE):** SSE measures how much our predicted values differ from the actual values in a squared manner. It's like calculating the sum of the squared "mistakes" our model makes in predicting the outcomes.

Formula:

$$SSE = \sum \left(Y_{actual} - Y_{predicted}\right)^2$$

Example :

• Suppose our model predicts student scores (Y_predicted) as 75, 80, 85, while the actual scores (Y_actual) are 80, 85, 90. SSE would be

$$(80-75)^2 + (85-80)^2 + (90-85)^2 = 25 + 25 + 25 = 75$$

• **The lower the SSE, the better our model is at predicting.**

Evaluating Linear Regression Model

2. **R-squared** is a good measure to evaluate the model fitness. It is also known as the coefficient of determination, or for multiple regression, the coefficient of multiple determination. The R-squared value lies between 0 to 1 (0%–100%) with a larger value representing a better fit. It is calculated as:

$$R^2 = 1 - \frac{SSE}{\sum\left(Y_{actual} - \overline{Y}\right)^2}$$

Evaluating Linear Regression Model

$$R^2 = 1 - \frac{\text{SSE}}{\Sigma(Y_{\text{actual}} - \bar{Y})^2}$$

Example Assume, We've built a LR model, and the predicted scores

Hours Studied (X)	Exam Scores (Y_actual)	Predicted Scores (Y_predicted)
2	60	65
3	70	75
4	80	85
5	85	95

Interpretation:
The calculated R-squared value of approximately **0.527 means that about 52.7%** of the variability in exam scores is explained by the number of hours studied. The remaining **47.3%** is not explained by our model.

1. **Total Variability ($\Sigma(Y_{\text{actual}} - \bar{Y})^2$):**

   $\bar{Y} = \frac{60+70+80+85}{4} = 73.75$

   $\Sigma(Y_{\text{actual}} - \bar{Y})^2 = (60 - 73.75)^2 + (70 - 73.75)^2 + (80 - 73.75)^2 + (85 - 73.75)^2$

   $\Sigma(Y_{\text{actual}} - \bar{Y})^2 = 222.5$

2. **Residual Variability (SSE):**

   $\text{SSE} = \Sigma(Y_{\text{actual}} - Y_{\text{predicted}})^2$

   $\text{SSE} = (60 - 65)^2 + (70 - 75)^2 + (80 - 85)^2 + (85 - 95)^2$

   $\text{SSE} = 105$

3. **Explained Variability ($\Sigma(Y_{\text{actual}} - \bar{Y})^2 - \text{SSE}$):**

   $\text{Explained Variability} = 222.5 - 105 = 117.5$

4. **Total Variability (R^2):**

   $R^2 = 1 - \frac{\text{SSE}}{\Sigma(Y_{\text{actual}} - \bar{Y})^2}$

   $R^2 = 1 - \frac{105}{222.5}$

   $R^2 \approx 0.527$

Example1 :Predicting Exam Scores

Suppose we want to predict students' exam scores (dependent variable) based on the number of hours they study (independent variable).

Hours Studied (X)	Exam Scores (Y)
2	60
3	70
4	80
5	85

Predicting Exam Scores .....

Step 1: Calculate the Mean of X and Y

$$\bar{X} = \frac{\Sigma X}{n}$$

$$\bar{Y} = \frac{\Sigma Y}{n}$$

$$\bar{X} = \frac{2+3+4+5}{4} = 3.5$$

$$\bar{Y} = \frac{60+70+80+85}{4} = 73.75$$

Hours Studied (X)	Exam Scores (Y)
2	60
3	70
4	80
5	85

Predicting Exam Scores

Step 1: Calculate Slope(b) and Intercept(a)

$$\mathbf{b} = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_i (X_i - \overline{X})^2}$$

$$\frac{(2-3.5)(60-73.75)+(3-3.5)(70-73.75)+(4-3.5)(80-73.75)+(5-3.5)(85-73.75)}{(2-3.5)^2+(3-3.5)^2+(4-3.5)^2+(5-3.5)^2} \boxed{= 8.5}$$

Hours Studied (X)	Exam Scores (Y)
2	60
3	70
4	80
5	85

$$a = (\overline{Y} - b\overline{X}) \boxed{= 44.00}$$

$$Y = (a + bX) + \varepsilon$$

So, the linear regression equation is Y=8.5X+44.

Predicting Exam Scores

Step 3: Predict Exam Score for a New Value of X

Let's say a student studies for **6 hours**.

$$Y_{predicted} = 8.5*6+44$$
$$= 95$$

Hours Studied (X)	Exam Scores (Y)
2	60
3	70
4	80
5	85

Predicting Exam Scores

Hours Studied (X)	Exam Scores (Y)
2	60
3	70
4	80
5	85

Step 4: Calculate Residual (Sum of Squared Errors)

Residual=$Y_{actual}-Y_{predicted}$

Residuals=[60−(8.5·2+44),70−(8.5·3+44),80−(8.5·4+44),85−(8.5·5+44)]

Residuals=[-1, 0.5, 2, -1.5]

$$SSE = \sum \left(Y_{actual} - Y_{predicted}\right)^2$$

Sum of Squared Errors (SSE)= $-1^2 + 0.5^2 + 2^2 + (-1.5)^2$
SSE=7.5
This represents the total error in our predictions.
The smaller the SSE, the better the model fits the data.

Example2:**Investigating Academic Performance**

A college professor is intrigued by the idea that there might be a correlation between students' grades in internal examinations and their subsequent performance in external examinations. To explore this hypothesis, the professor selects a random sample of 15 students from the class.

Exploring the Relationship:

- The professor is interested in understanding whether a high grade in internal exams tends to correlate with high grades in external exams.

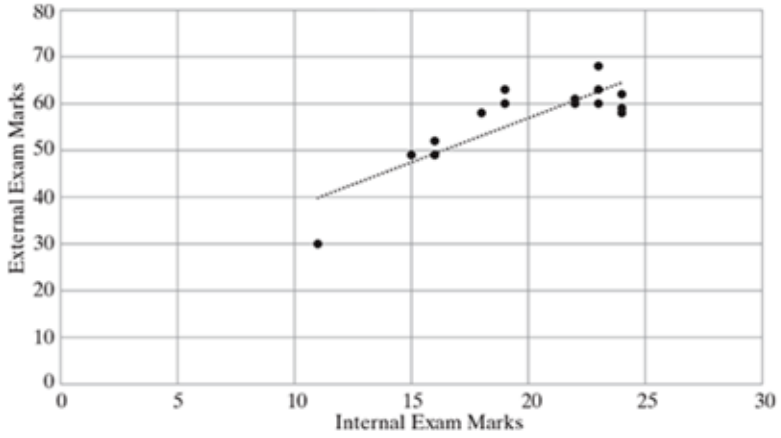
Data Collection:

- A random sample of 15 students is chosen for the study.
- The professor gathers data on the grades of these students in both internal and external examinations.

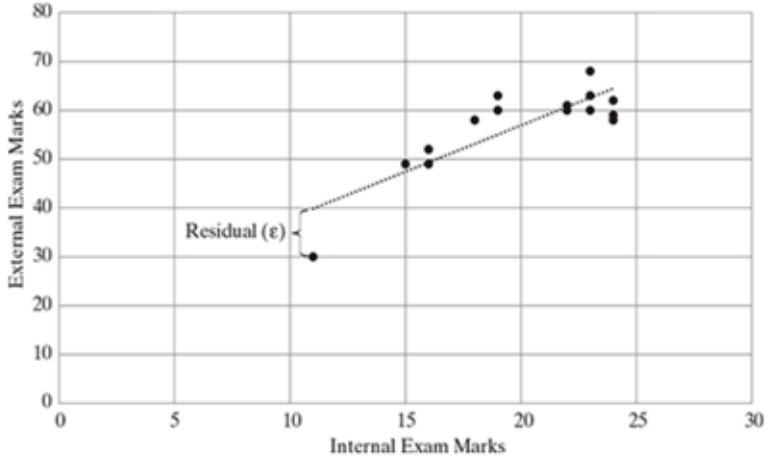
Internal Exam	15	23	18	23	24	22	22	19	19	16	24	11	24	16	23
External Exam	49	63	58	60	58	61	60	63	60	52	62	30	59	49	68

Investigating Academic Performance

As you can observe from the above graph, the line does not predict the data exactly. Instead, it just cuts through the data. Some prediction are lower than expected, while some others are higher than expected



Residual is the distance between the predicted point (on the regression line) and the actual point as depicted



Detailed calculation of regression parameters

X	Y	X- mean (X)	Y- Mean (Y)	$(Xi-\overline{X})(Yi-\overline{Y})$	$(Xi-\overline{X})^2$
15	49	-4.93	-7.8	38.454	24.3049
23	63	3.07	6.2	19.034	9.4249
18	58	-1.93	1.2	-2.316	3.7249
23	60	3.07	3.2	9.824	9.4249
24	58	4.07	1.2	4.884	16.5649
22	61	2.07	4.2	8.694	4.2849
22	60	2.07	3.2	6.624	4.2849
19	63	-0.93	6.2	-5.766	0.8649
19	60	-0.93	3.2	-2.976	0.8649
16	52	-3.93	-4.8	18.864	15.4449
24	62	4.07	5.2	21.164	16.5649
11	30	-8.93	-26.8	239.324	79.7449
24	59	4.07	2.2	8.954	16.5649
16	49	-3.93	-7.8	30.654	15.4449
23	68	3.07	11.2	34.384	9.4249
19.9	**56.8**		$\Sigma(Xi-\overline{X})(Yi-\overline{Y})$	**429.8**	**226.9335**

Step 1 Step 2 Step 3 Step 4 Step 5 Step 6

$$b = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_i (X_i - \overline{X})^2}$$

Step 7: Divide (step4 / step6)

b = 429.28 / 226.93 = 1.89

Step 8: Calculate a using the value of b

$a = \overline{Y} - b\overline{X}$

a = 56.8 – 1.89 × 19.9

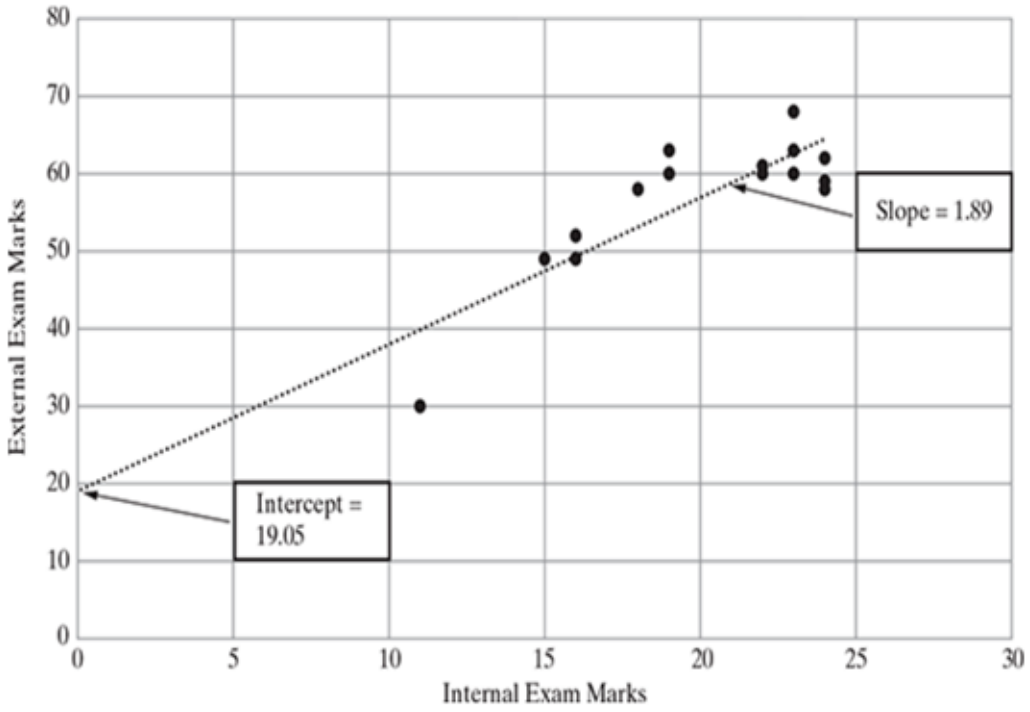
a = 19.05

The LR model of **Investigating Academic Performance**

Marks in external exam = 19.04 + 1.89 × (Marks in internal exam)

The **value of the intercept** from the above equation is 19.05.

Slope measures the estimated change in the average value of Y as a result of a one-unit change in X. Here, slope = 1.89 tells us that the average value of the external examination marks increases by 1.89 for each additional 1 mark in the internal examination.



Summary : Linear Regression

- When the outcome, or class, is numeric, and all the attributes are numeric, **linear regression** is a natural technique to consider.
- **Linear regression** is an excellent, simple method for numeric prediction, and it has been widely used in statistical applications for decades.

Non-linear Regression

- So far, we have seen the concept of simple linear regression where a single predictor variable X was used to model the response variable Y .
- In many applications, **there is more than one factor that influences the response**. **Multiple regression models** thus describe how a single response variable Y depends linearly on a number of predictor variables.
- A multiple linear regression model with k predictor variables X1, X2, ..., Xk and a response Y , can be written as
- $$\hat{Y} = a + b_1X_1 + b_2X_2 + b_3X_3 + \ldots + b_nX_n$$

Examples

- The selling price of a house can depend on the desirability of the **location**, the **number of bedrooms**, the **number of bathrooms,** the **year the house** was built, the square footage of the lot and a number of **other factors**.
- The height of a child can depend on the **height of the mother**, the **height of the father**, **nutrition**, and **environmental factors**.

Example :Predicting Exam Scores with Multiple Linear Regression

Interested in understanding what factors influence students' exam scores, a professor conducts a study at a university. The professor believes that the number of hours students dedicate to studying and the frequency of practice tests are essential.

Setting the stage:

- **Variables:**
  - *Y* (Exam Scores)
  - *X*1 (Hours Studied)
  - *X*2 (Practice Tests Taken)
- **Objective:**
  - Predict *Y* based on *X*1 and *X*2 using multiple linear regression.

Hours Studied (*X*1)	Practice Tests (*X*2)	Exam Scores (*Y*)
2	1	60
3	2	70
4	2	80
5	3	85

Example :Predicting Exam Scores with Multiple Linear Regression

Step 1: Set Up the Model:

Our multiple linear regression model is $Y = b_0 + b_1 \cdot X_1 + b_2 \cdot X_2$

Step 2: Calculate Means:

$$\bar{X}_1 = \frac{2+3+4+5}{4} = 3.5$$

$$\bar{X}_2 = \frac{1+2+2+3}{4} = 2$$

$$\bar{Y} = \frac{60+70+80+85}{4} = 73.75$$

Hours Studied (X1)	Practice Tests (X2)	Exam Scores (Y)
2	1	60
3	2	70
4	2	80
5	3	85

Step 3: Calculate Slopes (*b*1 and *b*2):

$$b_1 = \frac{(2-3.5)(60-73.75)+(3-3.5)(70-73.75)+(4-3.5)(80-73.75)+(5-3.5)(85-73.75)}{(2-3.5)^2+(3-3.5)^2+(4-3.5)^2+(5-3.5)^2}$$ =10.0000

$$b_2 = \frac{(1-2)(60-73.75)+(2-2)(70-73.75)+(2-2)(80-73.75)+(3-2)(85-73.75)}{(1-2)^2+(2-2)^2+(2-2)^2+(3-2)^2}$$ = -2.5000

Step 4: Calculate Intercept (*b*0):

$$b_0 = \bar{Y} - b_1 \cdot \bar{X}_1 - b_2 \cdot \bar{X}_2$$ =43.7500

The final multiple linear regression model would be:

Y= 43.75 + 10X1 - 2.5X2

Example :**Predicting Exam Scores with Multiple Linear Regression**

This model allows us to predict the exam score based on the number of hours studied and the number of practice tests taken . For example, if a student studies for 4 hours and takes 2 practice tests , the predicted exam score would be:

$$Y= 43.75+10*4 -2.5*2= 78.75$$

So, according to our multiple linear regression model, the predicted exam score would be 78.75 for a student who studies for 4 hours and takes 2 practice tests.

Linear Regression in Python

Importing Libraries

```
import pandas as pd
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
```

Loading Data into a DataFrame

```
# Given data in a DataFrame
data = {
    'Hours_Studied': [2, 3, 4, 5],
    'Exam_Scores': [60, 70, 80, 85]
}

df = pd.DataFrame(data)
```

Extracting Features and Target Variable

```
# Extracting features (X) and target variable (y)
X = df[['Hours_Studied']]
y = df['Exam_Scores']
```

Building and Fitting the Model

```
# Create and fit the linear regression model
model = LinearRegression()
model.fit(X, y)
```

Understanding and Display Coefficients

```
# Print the coefficients
print(f'Intercept (b0): {model.intercept_}')
print(f'Coefficient for Hours Studied (b1): {model.coef_[0]}')
```

Making Predictions on Existing Data

```
# Make predictions for the same data points
predicted_scores = model.predict(X)
```

Evaluating Model Performance

```
# Calculate SSE (Sum of Squared Errors)
sse = ((y - predicted_scores) ** 2).sum()
print(f'Sum of Squared Errors (SSE): {sse}')
```

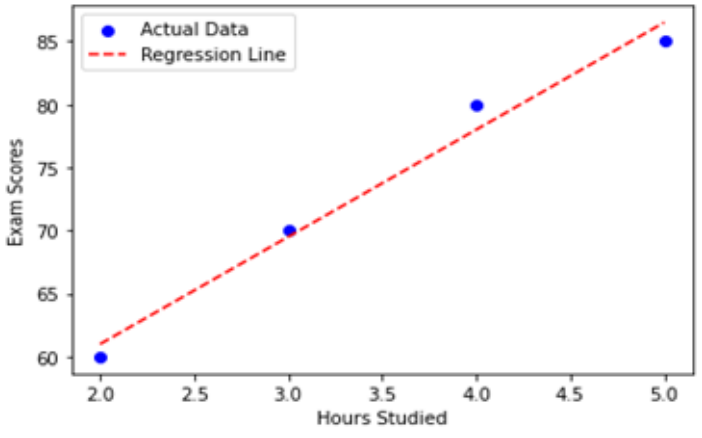
Linear Regression in Python

Making Predictions for New Data

```
# Make predictions for new data points
new_data = pd.DataFrame({'Hours_Studied': [6, 7, 8]})
predicted_scores = model.predict(new_data)
print('Predicted Scores:', predicted_scores)
```

Visualizing Results

```
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X, predicted_scores, color='red', linestyle='dashed', label='Regression Line')
plt.xlabel('Hours Studied')
plt.ylabel('Exam Scores')
plt.legend()
plt.show()
```



Result after playing the code

```
Intercept (b0): 44.000000000000014
Coefficient for Hours Studied (b1): 8.499999999999996
Sum of Squared Errors (SSE): 7.500000000000014
```